

Ivan Zolin

ML/AI Research Engineer

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SUMMARY

ML/AI Research Engineer working at the intersection of fundamental machine learning, LLMs/VLMs, NLP, and computer vision. Interested in new architectures, training methods, optimization, and AI systems.

EXPERIENCE

Digital Public Health Technologies (DPHT), ITMO University

NLP R&D Engineer

August 2025 – Present

Saint Petersburg, Russia

- Designed and developed MAS-EHR, a multi-agent system for automated processing of electronic health records, covering OCR, de-identification, and clinical information structuring.

Laboratory of Biomedical Imaging and Data Analysis (LABID), SPbPU

AI Researcher (Computer Vision & NLP)

April 2024 – April 2026

Saint Petersburg, Russia

- Developed TriDeFusion-v2, a hybrid RAUDen-based denoising architecture with Noise2Noise knowledge distillation and a CUDA wavelet-domain refinement module; manuscript under review in a Q1 journal.
- Developed NeuroRAG, a multi-agent LLM and RAG system over biomedical sources (PubMed, NCBI, arXiv); manuscript under review in a Q1 journal.

Laboratory of Molecular Neurodegeneration (LMN), SPbPU

Junior CV Researcher

April 2023 – April 2024

Saint Petersburg, Russia

- Developed TriDeFusion, a 3D fluorescence-microscopy denoising algorithm combining Noise2Noise and Non-local Means, presented at IEEE SIBIRCON 2024.
- Developed FRT, a web-based system for processing 3D microscopy images using machine learning and deep learning methods.

Partner Finder (startup)

ML Engineer

February 2023 – August 2023

Saint Petersburg, Russia

- Developed a search and recommendation system for business proposals, combining NLP-driven text representations with user-based features; scaled to 10k+ proposals.

Inream (Startup)

Kotlin Full-Stack Developer

November 2022 – December 2022

Saint Petersburg, Russia

- Developed a web interface for personalized language-learning tasks and deployed the system using Heroku.

EDUCATION

ITMO University

M.S., Deep Learning and Generative AI

GPA: 4.90/5.00, 3.82/4.00

Thesis: "Multi-agent Medical Records Data Processing System (EHR)"

August 2024 – June 2026

Peter the Great St. Petersburg Polytechnic University (SPbPU)

B.S., Applied Mathematics and Computer Science

GPA: 4.53/5.00, 3.62/4.00

Thesis: "Online AI-based deconvolution and denoising service for confocal microscopy"

September 2020 – June 2024

PROJECTS

MAS-EHR OCR

August 2025 – Present

- MAS-EHR: a multi-agent system for automated processing of electronic health records, covering OCR, de-identification, and structuring/extraction of clinical information.

[MAS-EHR Presentation](#)

NeuroRAG

April 2024 – Present

LLM, RAG, Multi-agent systems, Reciprocal Rank Fusion

- NeuroRAG: multi-agent LLM + RAG assistant for neurobiological and medical question answering over PubMed, NCBI and arXiv, combining Reciprocal Rank Fusion, LLM-based reranking and contextual compression with validated ensemble generation across several LLMs.
- Ranked first or tied for first on six of nine MMLU biomedical subsets (peak 99.0% on Medical Genetics) and was selected the best response 103 of 160 times (72.5% of decisive preferences, after excluding ties and inadequate responses) in a blind evaluation by 13 domain specialists.

[GitHub](#) • [Web service](#)

TriDeFusion

April 2023 – Present

Noise2Noise, Non-local Means, RAUDen, Wavelet Filtering, CUDA

- TriDeFusion (v1): 3D fluorescence-microscopy denoising algorithm combining Noise2Noise and Non-local Means; up to 54% RMSE reduction and 20% PSNR increase on synthetic data vs N2N-UNet (IEEE SIBIRCON 2024).
- TriDeFusion-v2: hybrid RAUDen architecture with Noise2Noise knowledge distillation (RAUDen-Lite, -40% parameters and inference time) and a CUDA wavelet-domain refinement module (8x speedup); under review in a Q1 journal.

[GitHub](#)

FRT

April 2023 – Present

FastAPI, Celery, Redis, S3, React, Docker

- Developed a web-based system for processing 3D microscopy images using ML and DL methods (FastAPI, Celery, Redis, S3, React, Docker).

[GitHub](#)

PUBLICATIONS

TriDeFusion-v2: Distilled Hybrid Residual Attention Denoising and Wavelet Filtering

Ivan Zolin, Anastasiya Rakovskaya, Evgenii Fedorov, Sergey Belyaev, Vyacheslav Chukanov, Ekaterina Pchitskaya 2026, IEEE Access

NeuroRAG: A Multi-Agent Model for Neurobiological Question Answering

Vladimir Skvortsov, Ivan Zolin, Vyacheslav Chukanov, Ekaterina Pchitskaya 2026, Journal of Biomedical Informatics

TriDeFusion: Enhanced denoising algorithm for 3D fluorescence microscopy images integrating modified Noise2Noise and Non-local means

Ivan Zolin, Vyacheslav Chukanov, Ekaterina Pchitskaya 2024, 2024 IEEE International Multi-Conference on Engineering, Computer and Information Sciences, SIBIRCON 2024

DOI: [10.1109/SIBIRCON63777.2024.10758532](https://doi.org/10.1109/SIBIRCON63777.2024.10758532)

[IEEE Xplore](#) • [SIBIRCON Proceedings PDF](#)

ACHIEVEMENTS

- **ODS Data Fest 2026** 2026: Speaker at ODS Data Fest 2026 ("Medical Data Horror Stories" track) with the talk "MAS-EHR: A Multi-Agent System for Clinical Data Integration and Harmonization".
- **International AI & Big Data Award "Gravitation 2026"** 2026: Winner, International AI & Big Data Award "Gravitation 2026" (Youth Track) for MAS-EHR.
- **XIV ITMO Congress of Young Scientists** 2025: Winner, XIV ITMO Congress of Young Scientists (2025).
- **International ITMO Conference "Young Professionals"** 2025: Participant, International ITMO Conference "Young Professionals" (2025).
- **Polytech for Life Sciences 2025** 2025: Conference presentation at "Polytech for Life Sciences 2025".
- **Open Data Forum 2025** 2025: Conference presentation at "Open Data Forum 2025".

- **IEEE SIBIRCON 2024** 2024: Presented the paper "TriDeFusion: Enhanced denoising algorithm for 3D fluorescence microscopy images integrating modified Noise2Noise and Non-local means" at IEEE SIBIRCON 2024.
- **International ITMO Conference "Young Professionals" 2024**: Winner, "Best Popular Science Talk" at the International ITMO Conference "Young Professionals" (2024).
- **Andrey Melnichenko Russian Science Competition 2024**: Grand Finalist, Andrey Melnichenko Russian Science Competition (2024).
- **Blue Sky Research Scientific Projects Competition 2024**: Participant, Blue Sky Research Scientific Projects Competition (2024).
- **Medicine at the Intersection of Modern Technologies 2024**: Conference presentation at "Medicine at the Intersection of Modern Technologies" (2024).
- **Open Data Forum 2024** 2024: Team participant, "Open Data Forum 2024".
- **Selectel Career Wave Scholarship 2024**: Recipient of the Selectel Career Wave Scholarship (Autumn 2024).
- **ITMO Students Awards 2024** 2024: Nominee for the "Intellect of the Year" award at ITMO Students Awards 2024.
- **Polytech for Life Sciences 2023** 2023: Conference presentation at "Polytech for Life Sciences 2023".
- **CB&AI – Computational Biology and Artificial Intelligence for Personalized Medicine 2023** 2023: Conference presentation (thesis), "CB&AI – Computational Biology and Artificial Intelligence for Personalized Medicine" (2023).
- **Pitch the Future: Med AI**: Recipient of the Selectel grant for the "Pitch the Future: Med AI" competition, with the MAS-EHR project.
- **Detection and Anonymization of Personal Data in Unstructured Medical Texts**: Registered patent for a system detecting and anonymizing personal data in unstructured medical texts.
- **AI System Based on an Ensemble of Language Models with RAG Access to a Neuroscience Knowledge Base**: Registered patent for an AI system based on an ensemble of language models with RAG-based access to a neuroscience knowledge base.
- **Automatic Segmentation of Fluorescent Microspheres in Confocal Microscopy Images**: Registered patent for a method of automatic segmentation of fluorescent microspheres in confocal microscopy images.
- **Online AI-Based Deconvolution and Denoising Service for Confocal Fluorescence Microscopy Images**: Registered patent for an online AI-based deconvolution and denoising service for confocal fluorescence microscopy images.

SKILLS

Languages: Python, C, C++, SQL, MATLAB, JavaScript, Java, LaTeX

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, LLM, RAG, Multi-Agent Systems, RL, AutoML

Frameworks: PyTorch, JAX, TensorFlow, OpenCV, ONNX, LangChain, LangGraph, scikit-learn, Ultralytics, pandas, NumPy, NLTK

Data: PostgreSQL

MLOps: Docker, FastAPI, Django, ReactJS, Redis, Kafka, Celery, S3, Prometheus, Grafana, ELK, CI/CD, Microservices, GPU/CUDA

Core CS: Statistics, Probability Theory, Calculus, Linear Algebra, Optimization, Discrete Mathematics, Algorithms and Data Structures, Parallel Computing

Tools: Git, Bash